# PENGWIN 2026 Task 1 — V1.3.3 STU-Net Two-Stage Submission

Algorithm Description

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> **V1.3.3** is an inference-only refinement on top of V1.3.2 (the model is **unchanged**). V1.3.2 took the leaderboard from ~0 to **fracture Dice 0.767** by fixing a random-weight bug (below); V1.3.3 then targets the dominant remaining error revealed by the official per-case metrics — **under-segmentation** (instance recall 0.456, merge_error 0.65, split_error 0). A multi-agent code audit with adversarial verification found the cause: the deploy decoder's non-Sacrum **size-ratio merge** (0.05) was collapsing genuine 1–11 cm³ fracture fragments — and had silently **diverged** from the validated offline evaluator, which uses 0.0. Disabling it (Sacrum keeps its deliberate 0.10 precision guard) recovers the real fragments: on a local instance-level proxy, mean recall **0.46 → 0.51**, F1 **0.50 → 0.55**, with **split_error still 0 and precision held**. V1.3.3 also removes a speckle-era "bbox > 50 %" band-aid that fired on *clean* large anatomy in a tight FOV (swapping the soft probability ROI for a hard bone mask → out-of-distribution decode), and deletes ~120 lines of dead/legacy code. A confidence-based routing guard for the cross-group hallucination was prototyped and **rejected on the evidence** — Ds539 is *confidently wrong* (real and phantom bones both score 0.97–0.99 softmax), so that is a model-level issue for future retraining, not an inference threshold.

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> **V1.3.2** fixes the *actual* root cause of the earlier ~0 Grand-Challenge scores and keeps the V1 STU-Net model and I/O contract unchanged. The GC container pins `nnunetv2==2.5.1`, whose `initialize_from_trained_model_folder()` builds the network but does **not** load the checkpoint into `predictor.network` (it defers the load to `perform_actual_prediction()`). Our custom inference path reads `predictor.network` directly and never triggers that deferred load, so **both stages ran with random initialization** → speckle anatomy (tens of thousands of connected components) → ~0. The local dev env happened to run nnUNet **2.5.2**, which *does* load at init (MIC-DKFZ/nnUNet#2520), so the bug was invisible locally. V1.3.2 loads the trained weights into the network **explicitly** in `build_predictor`, making inference correct on any nnUNet version. It also retains the V1.3 robust per-anatomy routing and the self-diagnostic logging (now including a per-stage weight checksum `w0sum`). **Model weights are identical to V1.**

# 0. Submission intent

Per-fragment instance segmentation of pelvic and femoral bone fractures (sacrum, left/right hip, femur) in CT. A **STU-Net two-stage anatomy-conditioned pipeline, femur fully modeled**, trained on a leakage-free, case-grouped split. Supersedes the V0.3.x pipeline-test (ResEnc, pelvic-only, femur-stub).

# 1. Method overview

A two-stage, anatomy-conditioned cascade wrapped in a robustness shell.

- **Stage 1 — anatomy** (`Dataset539_PelvicFemurAnatomyV3`): a whole-CT 5-class semantic model (`0=bg, 1=sacrum, 2=leftHip, 3=rightHip, 4=femur`).
- **Per-anatomy ROI**: for each present bone, take the bbox of the Stage-1 probability `>= 0.5` (padded), and build a **3-channel ROI** — bone-LUT CT, Stage-1 anatomy probability, and a signed distance map clipped to +/-40 mm.
- **Stage 2 — fracture** (`Dataset538_PelvicFemurBICMFragmentV5`): a per-anatomy ABBC model that emits a 4-class field (`background / border / boundary / core`).
- **Decoder**: core-seed watershed — connected components of the core class are watershed seeds; support is flooded from them; a >=1 cm^3 component prune and a small-fragment merge follow.
- **Assembly**: per-bone fragment IDs are offset into the official PENGWIN ranges (sacrum 1-50, leftHip 51-100, rightHip 101-150, **femur 151-200**) and the volume is reoriented to the input frame.

**Backbone.** Both stages use **STU-Net-B** (58 M params, Apache-2.0) warm-started from a TotalSegmentator bone pretrain — a license-clean transfer that initializes the encoder for pelvic/femoral bone.

# 2. Stage 1 — Ds539 anatomy

- STU-Net-B, `nnUNetResEncUNetLPlans` / `3d_fullres`, 1-channel CT input, 5-class softmax. Trainer `PengwinTrainerSTUNetBaseAnatomyV301`.
- Input: LPS canonicalize + raw HU fed through nnUNet's own CTNormalization (foreground-percentile clip + z-score) and resampling to the model's target spacing — the standard nnUNet preprocessing the model was trained with.
- Output: full-CT 5-class anatomy probability, resampled back to the CT grid.

# 3. Per-anatomy routing (robust — no pelvic/femur gate) [changed in V1.3]

Ds539's marginal training hallucinates the cross-group anatomy: a pelvic case gets a phantom femur channel, and a femur case gets phantom pelvic channels. V1 gated the **whole case** to "pelvic" or "femur" by a single femur/pelvic argmax volume ratio (threshold 0.45); a borderline hallucination flipped the ratio and routed the case to the **wrong** anatomy set, scoring it **0**. End-to-end testing showed this misrouted ~25 % of cases (both directions).

V1.3 removes the gate: it processes **every anatomy whose Ds539 argmax mask is at least 20 % of the largest present mask**. The genuinely-present bone(s) are thus kept every time — a small hallucination is dropped by the fraction gate, a sizable one becomes a minor false-positive, never a zero. End-to-end batch (8 cases): **mean fracture Dice 0.726 → 0.968, zero 0-score cases**.

# 4. Stage 2 — Ds538 ABBC fracture

- STU-Net-B, 3-channel input, 4-class ABBC head. Trainer `...DenseCandidateCore025StrongPeakNoContactABBCSTUNetBV301` (a V300 boundary-attention refinement). Aligned with the PENGWIN 2024 1st-place (MIC-DKFZ) ABBC formulation: explicit contact-voxel classification is dropped in favour of a core-seed -> boundary watershed.
- Training: unified **case-grouped** 5-fold split (seed 12345) shared with Stage 1, so a held-out fold is held out across *both* stages (no patient leakage).

# 5. Core-seed watershed decoder

Core class -> `scipy.ndimage.label` seeds; watershed flood over the support mask; >=1 cm^3 connected-component prune; small-fragment merge. An anatomy-specific over-segmentation control merges the sacrum's spurious core islands (Sacrum Instance-F1 0.585 -> 0.892) while the multi-fragment hips/femur keep the defaults.

# 6. Robustness shell

L1) bone-skeleton anatomy decomposition (HU>200 + 3D CC) — distribution-independent fallback. L2) Ds539 argmax masks + the V1.3 multi-anatomy routing above. L3) post-pad bbox sanity (<= 50 % of volume) — rejects OOD masks. L4) 480 s time budget — guarantees the 10-minute GC limit. Largest-CC-keep per anatomy.

# 7. Self-diagnostic logging [V1.3.1; w0sum diagnosis confirmed in V1.3.2]

The container logs, every run: the **loaded network class** (STU-Net vs the plans' ResEnc) plus a **weight checksum** `w0sum` (the abs-sum of the stem conv); the **input raw-HU distribution** (min/max/mean/percentiles, bone>200 and air<-500 fractions); and **each Ds539 anatomy's volume fraction with a GARBAGE flag**. This logging is what pinned the V1.3.2 root cause: `w0sum` was **different on every GC case** (≈80.5, 81.7) instead of the trained checkpoint's constant **104.03** — the signature of an *unloaded, randomly-initialised* network (see the header). The fix restores a constant `w0sum=104.03` (Ds539) / `194.38` (Ds538) on the GC container, so a 0-score run is self-explaining from the log.

# 8. Performance (held-out, dev proxy)

Held-out grouped fold-0 val (n=132 per-anatomy ROIs), scored with our PENGWIN-2026 **official-aligned v2 proxy** (per-anatomy argmax IoU>=0.10):

| metric | overall | Femur | RightHip | LeftHip | Sacrum |
|---|---|---|---|---|---|
| Fracture Dice | **0.799** | 0.845 | 0.854 | 0.759 | 0.732 |
| Instance F1 | **0.919** | 0.953 | 0.950 | 0.876 | 0.892 |
| HD95 (mm) | **18.9** | 7.4 | 25.7 | 25.7 | 18.3 |

**Scope of this proxy.** It scores **Stage 2 on GT-derived ROIs** (oracle Stage-1 anatomy), so it measures the fracture decoder, not the full cascade. True end-to-end additionally depends on Stage-1 anatomy + the V1.3 routing (section 3); an end-to-end batch over full cases reaches mean fracture Dice ~0.97 when routed correctly. The official Grand-Challenge evaluator is unpublished; these are aligned proxies, not leaderboard results.

# 9. Hardware + runtime

NVIDIA T4 16 GiB; STU-Net-B inference ~4 GiB/stage (serial load, peak = max of the two stages); per-case 45-205 s, capped at 480 s. Container: PyTorch 2.1.2 + CUDA 11.8 + nnUNetv2 2.5.1.

# 10. Limitations

- **Dev measurement only** — the official test set is unreleased.

- The held-out proxy is Stage-2-on-oracle-ROIs (section 8); end-to-end is gated by Stage-1 anatomy + routing.
- Single fold, single seed; no ensembling.

# 11. References


- nnU-Net v2: Isensee, F. et al. *Nat Methods* 18, 203-211 (2021).
- STU-Net (TotalSegmentator bone pretrain).
- PENGWIN 2026 Task 1 baseline: github.com/YzzLiu/PENGWIN2026_Task1_AutoSeg_Baseline
- PENGWIN 2024 1st place (ABBC reference): MIC-DKFZ.